Experiment No.: 3

Web and Connectivity

AIM

To practice effective web searching using advanced search operators, understand email etiquette, and familiarize students with Google Classroom.

REQUIREMENTS

- Computer or laptop with internet connectivity
- Web browser such as Google Chrome or Mozilla Firefox
- Search engine
- Email account
- Google account and access to Google Classroom

THEORY

1. Web Connectivity

Web connectivity refers to the ability of a computer or other device to connect to the internet and access online resources and services.

2. Advanced Search Operators

Search operators are special words or symbols used to refine a search and obtain more relevant results.

lightblue **Search operator**	**Purpose**
`"Computer Networks"`	Searches for the exact phrase "Computer Networks".
`site:edu computer engineering`	Searches for computer engineering information on educational websites using the `site:` operator.
`filetype:pdf Linux notes`	Searches for PDF documents related to Linux notes.
`Python OR Java`	Searches for results containing Python or Java.
`Java -coffee`	Searches for Java while excluding results containing the word coffee.

3. Email Etiquette

Email etiquette refers to the accepted practices for writing clear, polite, and professional emails. A formal email should have a suitable recipient, a meaningful subject, a polite message, and an appropriate closing.

4. Google Classroom

Google Classroom is an online learning platform used to access learning materials, join classes, and submit assignments.

PROCEDURE

Activity 1: Advanced Search Operators

1. Open a web browser.
2. Open a search engine.
3. Enter the following queries one by one:
   - `"Computer Networks"`
   - `site:edu computer engineering`
   - `filetype:pdf Linux notes`
   - `Python OR Java`
   - `Java -coffee`
4. Observe the search results for each query.
5. Compare the results and note how each operator changes the search.

Activity 2: Email Etiquette

1. Open an email application and sign in.
2. Compose a new email.
3. Enter the recipient's email address.
4. Enter a clear and meaningful subject.
5. Write a polite message using appropriate language.
6. Attach a file if required.
7. Verify the recipient, subject, message, and attachment before sending.
8. Send the email and verify its status.

Activity 3: Google Classroom

1. Open Google Classroom.
2. Sign in using the permitted Google account.

3. Join a class using the class code provided by the instructor, if required.
4. Open the class and access the learning materials.
5. Open a sample assignment and submit it if submission access is available.
6. Verify whether the submission was successful.

OUTPUT / OBSERVATIONS

1. Record the results obtained for the five search queries.
2. Record one difference observed between an ordinary search and a search using an operator.
3. Verify whether the formal email was sent successfully.
4. Record whether the class materials were accessed and the sample assignment was submitted successfully.

Observation 1: Search results

Observation 2: Email status

Observation 3: Google Classroom access

RESULT

Thus, advanced search operators, email etiquette, and Google Classroom functionalities were studied and practised successfully.

PRACTICAL RELATED QUESTIONS

1. What is a search operator?
2. What is the purpose of the `site:` operator?
3. What is the use of the `filetype:` operator?
4. Why is email etiquette important?
5. What is Google Classroom?
6. Name any two web browsers.

SELF-LEARNING ACTIVITY

Use the query `filetype:pdf "Computer Engineering" site:edu` in Google. Record the URL and file size of the first PDF result found.

Experiment No.: 4

Professional Documentation

AIM

To create a formal report using a word processing application by applying document styles, page numbering, and an automatically generated table of contents.

REQUIREMENTS

- Computer or laptop
- Word processing application such as LibreOffice Writer or Microsoft Word
- Keyboard and mouse
- A suitable topic and the information required for the report

THEORY

1. Professional Documentation

Professional documentation is the process of preparing documents in a clear, organized, readable, and consistent format for academic or professional purposes.

2. Document Styles

Styles are predefined formatting settings that can be applied consistently to text.

lightblue **Style**	**Purpose**
Title	Formats the main title of the document.
Heading 1	Formats the main section headings.
Heading 2	Formats the subsection headings.
Normal	Formats the regular body text.

Using styles maintains consistency and helps the word processor identify headings when generating a table of contents.

3. Table of Contents (TOC)

A table of contents lists the major headings and subheadings of a document, usually with their page numbers. When heading styles are applied correctly, it can be generated automatically.

4. Page Numbering

Page numbering identifies the pages of a document and helps readers locate specific sections.

PROCEDURE

Activity: Creating a Formal Report

1. Open LibreOffice Writer or another available word processing application.
2. Create a new document.
3. Enter the report title: **Computer Fundamentals Report**.
4. Organize the report using the following structure:
    1. Introduction
    2. Hardware Components
    3. Software Components
    4. Advantages of Computers
    5. Applications of Computers
    6. Conclusion
5. Enter suitable content under each heading.
6. Select the main report title and apply the **Title** style.
7. Apply **Heading 1** to the main section headings.
8. Apply **Heading 2** to any subheadings.
9. Use the **Normal** style for body text.
10. Check the font, alignment, spacing, and overall readability of the report.
11. Insert page numbers using the available page-numbering feature.
12. Place the cursor where the table of contents should appear.
13. Use the application's automatic table of contents feature to generate the TOC from the heading styles.
14. Save the document with a suitable file name.
15. Reopen or preview the document and verify the headings, page numbers, and table of contents.

OUTPUT / OBSERVATIONS

1. Attach a screenshot of the completed report.
2. Attach a screenshot showing the applied heading styles.
3. Attach a screenshot of the automatically generated table of contents.

4. Verify that page numbers are displayed correctly.
5. Record the file name and format used to save the report.

Observation 1: Document formatting

Observation 2: Table of contents

Observation 3: Page numbering and saved file

RESULT

Thus, a formal report was created successfully using document styles, page numbering, and an automatically generated table of contents.

PRACTICAL RELATED QUESTIONS

1. What is professional documentation?
2. What is a document style?
3. Why are styles used in a report?
4. What is a table of contents?
5. Which style is used for main headings?
6. Name any two word processing applications.
7. Why is consistent formatting important?
8. How can an automatic table of contents be generated?
9. What is the purpose of page numbering?
10. State one advantage of professional documentation.

SELF-LEARNING ACTIVITY

Create a one-page assignment cover sheet using appropriate document styles. Then prepare a professional biodata for an internship application and export it as a PDF.

Experiment No.: 9

Data Analysis

AIM

To analyse a dataset using Pivot Tables and Slicers in Google Sheets for summarizing and filtering information.

REQUIREMENTS

- Computer or laptop with internet connectivity
- Web browser
- Google account
- Google Sheets
- A sample dataset containing suitable column headings and records

THEORY

1. Data Analysis

Data analysis is the process of organizing and interpreting data to extract useful information.

It is used in student performance analysis, attendance analysis, sales analysis, survey reports, and budget planning.

2. Pivot Table

A Pivot Table is a spreadsheet feature used to summarize and rearrange large amounts of data dynamically.

Its functions include:

- Calculating totals
- Calculating averages
- Counting records
- Grouping data
- Rearranging information

Pivot Tables help users analyse data more efficiently and generate organized reports.

3. Slicer

A Slicer is a visual filtering tool used with Pivot Tables. It allows users to filter data interactively by selecting categories.

Its advantages include:

- Interactive filtering
- Easy selection of categories
- Improved understanding of data
- Faster exploration of summarized information

PROCEDURE

Activity: Summarizing and Filtering Data

1. Open a web browser and sign in to Google Sheets using an authorized Google account.
2. Create a new spreadsheet or open a suitable dataset provided by the instructor.
3. Ensure that the dataset contains meaningful column headings and records.
4. Check the data for missing headings, blank rows, and inconsistent entries.
5. Select the dataset that will be analysed.
6. Open the Pivot Table feature and create a Pivot Table using the available spreadsheet options.
7. Choose the appropriate source data range and confirm the Pivot Table creation.
8. Configure the row, column, and value fields according to the information to be summarized.
9. Select a suitable summary operation, such as SUM, AVERAGE, or COUNT, where appropriate.
10. Examine the resulting Pivot Table and verify whether the data has been summarized as intended.
11. Add a Slicer using the available spreadsheet options and select the relevant column for filtering.
12. Apply different filter selections through the Slicer.
13. Observe how the displayed analysis changes when different categories are selected.
14. Clear or change the filter and verify that the displayed information updates appropriately.
15. Save the spreadsheet and verify the final output.

OUTPUT / OBSERVATIONS

1. Attach a screenshot of the original dataset.
2. Attach a screenshot of the completed Pivot Table.

3. Attach a screenshot showing the Slicer and the selected filter.
4. Record the fields used in the Pivot Table.
5. Record the summary operation applied to the values.
6. Describe how the displayed data changed after applying a filter.
7. Verify that the spreadsheet was saved successfully.

Observation 1: Dataset and selected fields

Observation 2: Pivot Table summary

Observation 3: Slicer filtering

RESULT

Thus, a dataset was analysed using a Pivot Table and a Slicer in Google Sheets to summarize and filter information.

PRACTICAL RELATED QUESTIONS

1. What is data analysis?
2. What is a Pivot Table?
3. State any two functions of a Pivot Table.
4. What is a Slicer?
5. How does a Slicer help in data analysis?
6. Name any two applications of data analysis.
7. What is the purpose of the value field in a Pivot Table?
8. Mention any two summary operations that can be used in a Pivot Table.

SELF-LEARNING ACTIVITY

Create a spreadsheet containing sample student records. Use a Pivot Table to summarize the records by a suitable category. Add a Slicer and demonstrate how the summary changes when a different category is selected.